# Security Architecture for Internet of Things (IoT) Environments

## Applying Defence-in-Depth Principles to Secure Connected Devices and Critical Infrastructure

### Security Architecture Design Study

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# Security Design Objectives

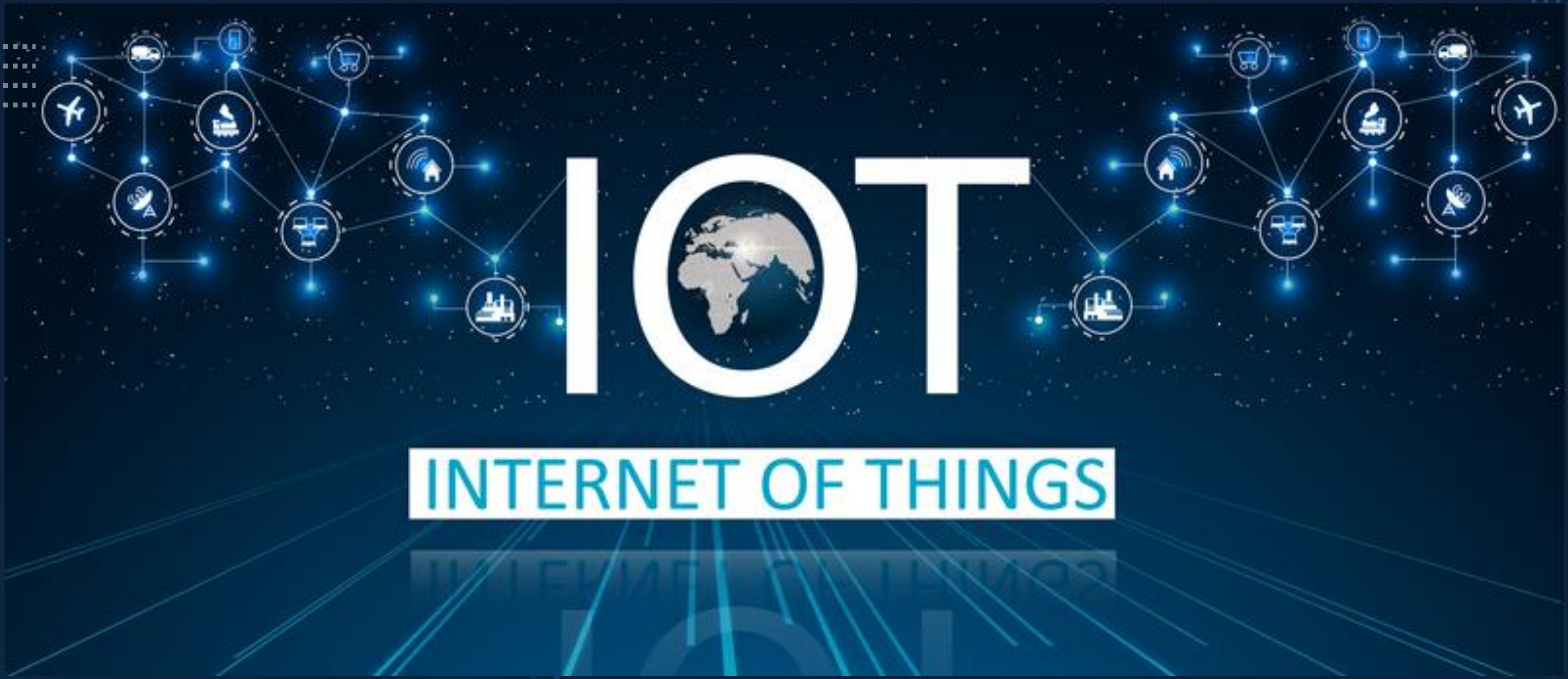
Improve
- Improve resilience of connected IoT environments.

Apply
- Apply defence-in-depth across device, network, and data layers.

Reduce
- Reduce attack surfaces through layered security controls.

Strengthen
- Strengthen authentication, encryption, and monitoring.

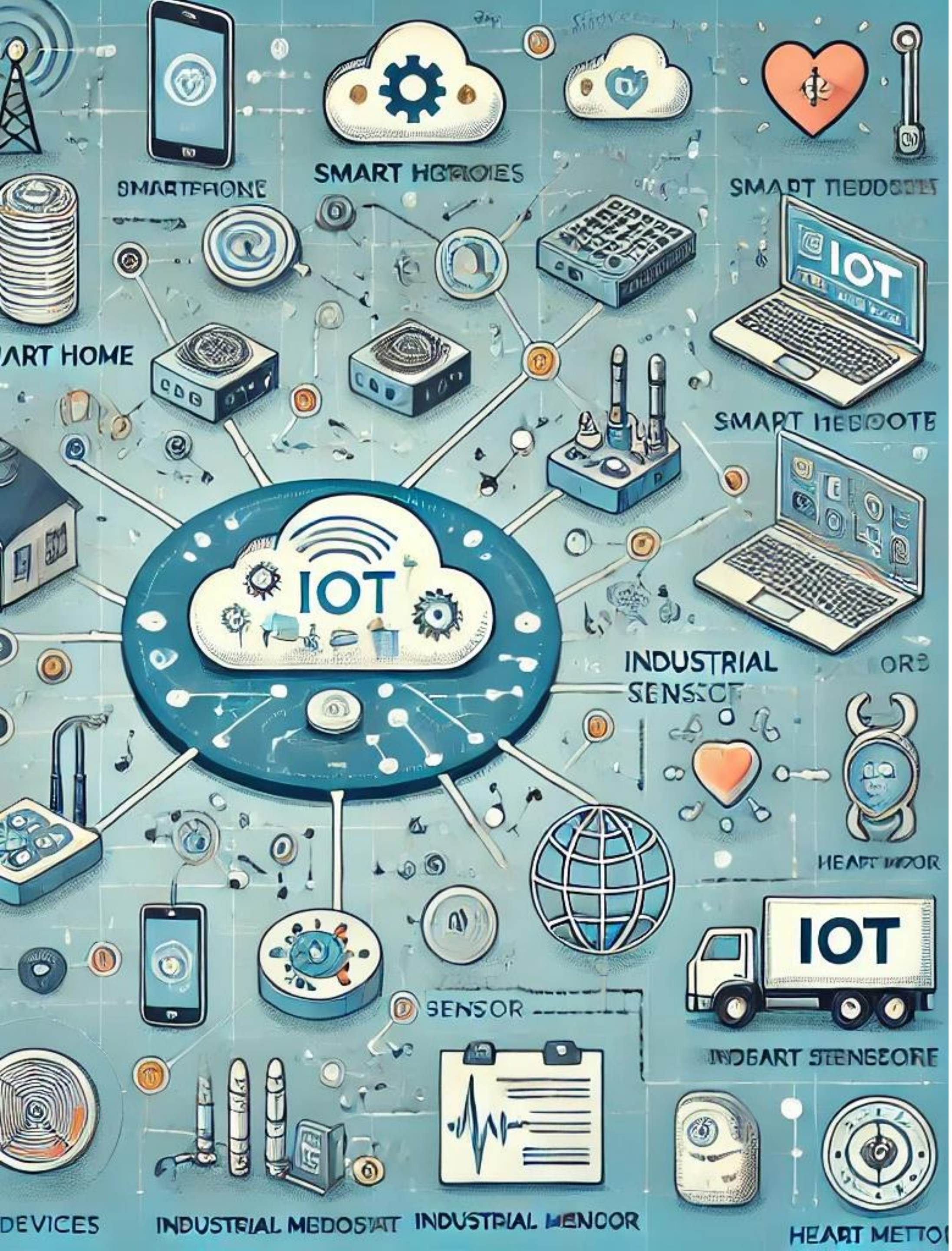
Evaluate
- Evaluate mitigation strategies against representative IoT threats.




# IoT Overview

**What is IoT?**

- A network of interconnected devices that collect, transmit, and process data without human intervention.
- Over 15 billion connected IoT devices globally as of 2023, expected to increase exponentially.




# IoT Ecosystem Overview

An illustration showing various interconnected IoT devices and systems.


# Cybersecurity Challenges in IoT

- IoT devices introduce significant cybersecurity risks.
- Challenges include resource constraints, lack of robust security measures, and decentralized nature.
- Vulnerabilities to data breaches, malware, and physical attacks.


# IoT Threat Landscape

**Types of IoT Devices**:

- Smartphones
- Smart Home Devices
- Wearable Devices Industrial IoT
- Medical Devices.

# Representative Attack Scenarios

**Botnets**: Malware enables attackers to access IoT devices and infiltrate networks.

**Weak Passwords**: Default or simple passwords make devices easy targets.

**Data Theft**: Attackers steal sensitive financial or personal information.

**DoS/DDoS Attacks**: Overloading devices or networks with traffic disrupts services.

# Impact of IoT Cybersecurity Threats

**On Individuals**: Identity theft, invasion of privacy, and physical harm.

**On Organizations**: Operational disruptions, data loss, financial losses, and reputational damage.

**On Critical Infrastructure**: Power outages, compromised public safety.

# CASE STUDIES

**The Mirai botnet:** Mirai is a botnet malware that exploits poorly protected IoT devices. In 2016, it infected IoT devices by leveraging weak credentials (e.g., default passwords) to build a network of compromised devices. This launched massive DDoS attacks, disrupting major websites like Amazon, Twitter, and Netflix.

**The Meris botnet**: Meris is an advanced botnet malware, an augmented and refined version of Mirai. It exploits vulnerabilities in IoT devices, particularly targeting routers with outdated firmware, to launch powerful DDoS attacks, peaking at 21.8 million requests per second (RPS).

# Strategies for Mitigating Cybersecurity Threats In IoT Devices

**Device-Level Security**: Strong authentication, encryption, firmware updates, physical hardening.

**Network-Level Security**: Network segmentation, firewalls, intrusion detection systems (IDS).

**Data Security**: Data minimization, tokenization, regular backups, IoT Security by Design, and Legal/Regulatory Approaches.

# Security Design Recommendations

**Automate**

- **Regular Updates**: Automate firmware and software updates to address vulnerabilities.

**Implement**

- **Device Identity Management**: Implement strong authentication and authorization.

**Encrypt**

- **End-to-End Encryption**: Encrypt data across all IoT communications.

**Use**

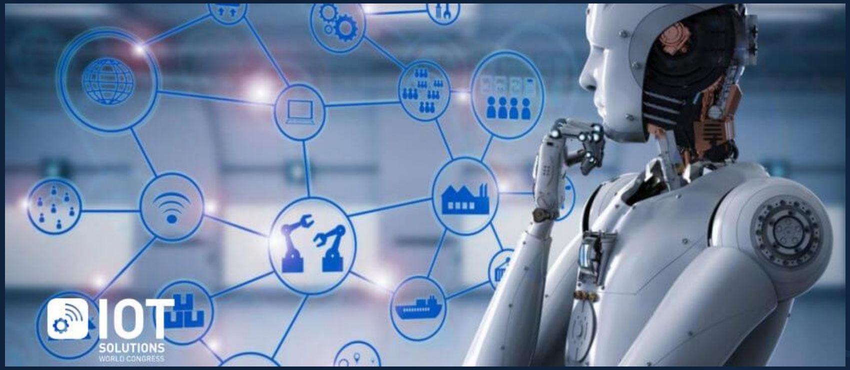
- **AI and ML Security**: Use AI and ML for advanced threat detection and response.

**Training**

- **User Training**: Enhance security awareness through targeted training programs.

# Conclusion

Securing Internet of Things (IoT) environments requires a defence-in-depth approach that combines secure device configuration, network segmentation, strong authentication, encryption, continuous monitoring, and timely patch management. As IoT adoption continues to grow across enterprise and critical infrastructure environments, applying recognised security standards and layered defensive controls remains essential to improving cyber resilience.



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